SWIFTCART ONLINE RETAIL ANALYSIS

Data Quality & Validation Report

Sprint 1 - Data Understanding, Cleaning and Validation

1. Executive Summary

This report summarizes the data checks carried out while preparing the SwiftCart dataset for exploratory data analysis. The work covered missing values, duplicate records, data types, invalid values, category consistency, date fields, postal codes and potential outliers. The aim was to make the dataset consistent and usable for analysis without removing records or changing values without a clear reason.

31,047	21	0	2,084
Rows	Columns	Exact duplicate rows in final data	Remaining missing cells

Overall result

The cleaned dataset is suitable for Sprint 2 EDA, with documented limitations. The final data contains no exact duplicate rows, key numeric fields have valid ranges, and category labels have been standardized. A small number of missing values remain where the original information could not be recovered reliably.

2. Dataset at a Glance

Item	Final result	Validation note
Dataset shape	31,047 rows x 21 columns	Final cleaned shape reported in the notebook outputs.
Exact duplicates	0	No exact duplicate rows remained in the final dataset.
Date coverage	30,559 valid dates; 488 missing	Invalid or unparseable dates were kept as missing rather than guessed.
Numeric ranges	Age 18-78; quantity 1-20; amount Rs. 229-Rs. 3,036	No zero or negative quantity or amount values were found.
Currency / country	INR / IN	All 31,047 records use INR and country code IN.

Scope note: This report is based on the cleaning and validation outputs shared during the project. A before-cleaning exact-duplicate count and a formal end-to-end accuracy check against the original business system were not available.

3. Missing Value Assessment

Missing values were reviewed column by column. Values were filled only where the project had a reasonable basis to do so. Identifiers and dates were not invented when the original value could not be reliably recovered.

Column	Before cleaning	After cleaning	What was done
age	1,085 missing; invalid ages above 100 also identified	0 missing	Converted to numeric, treated ages above 100 as invalid, then filled remaining missing ages with median 37. Final range: 18-78.
gender	1,075	0 missing	Missing values labeled as Unknown.
age_group	1,068	0 missing	Filled using age-related mode logic based on the available dataset pattern.
customer_id	1,079	41 missing	Recovered where possible using available order-ID information; 41 remained missing.
order_id	1,080	1,080 missing	Left missing because a reliable replacement ID could not be derived.
date	465	488 missing	Converted Excel serial dates and parsed dates; 23 additional malformed or unparseable values became missing.
month	465	465 missing	Standardized month labels; missing month values were left missing.
ship_postal_code	0 reported initially	10 missing	Ten values recorded as 12500700 did not match the expected six-digit format and were set to missing.

Final missing values

The final dataset has 2,084 missing cells across five columns: order_id (1,080), customer_id (41), date (488), month (465), and ship_postal_code (10). This is approximately 0.32% of all 651,987 cells in the 31,047 x 21 dataset.

Why some values remain missing

The remaining gaps are mainly identifiers and date fields. Filling them with guesses could create incorrect customer/order links or misleading time-based analysis. These gaps are documented and should be considered when interpreting results.

4. Data Type, Format and Range Checks

The project checked whether important columns were stored in a suitable format and whether their values were within reasonable ranges for this dataset.

Field / group	Final format or result	Validation performed
date	datetime64[ns]	Parsed to datetime for monthly and date-based analysis; 488 unavailable dates remain missing.
age	Integer; 18-78	Values above 100 were treated as invalid; remaining gaps filled using median 37.
quantity	Integer; 1-20	No missing, zero or negative values in the final dataset.
amount	Integer; Rs. 229-Rs. 3,036	No missing, zero or negative values in the final dataset.
ship_postal_code	Nullable Int64 in notebook	31,037 valid non-missing entries; 10 malformed eight-digit entries were set to missing.
b2b	Boolean	Final values: False 30,861; True 186.
currency / ship_country	Categorical text	All records show INR and IN respectively.
index	Integer; 1-31,047	Recreated as a sequential unique index; no missing index values in the final data.

5. Category and Text Consistency

Columns	Issue / check	Action and final result
status, channel, category, month	Extra spaces and inconsistent capitalization / labels	Whitespace removed and labels standardized (including kurta to Kurta).
ship_city	Inconsistent spacing and capitalization	Trimmed and title-cased; 2,603 unique city labels reported.
ship_state	Inconsistent state labels; New Delhi represented separately	Labels standardized and NEW DELHI mapped to DELHI; distinct labels reduced from 50 to 35.
size	Size labels checked	11 observed values retained: XS, S, M, L, XL, XXL, 3XL, 4XL, 5XL, 6XL and Free.
sku	Unwanted surrounding whitespace	Whitespace stripped; repeated SKUs retained because they represent repeated product records.

CSV data type note

CSV files do not store Pandas data types. When the cleaned CSV is reloaded, the date column should be parsed again. The nullable postal-code column may load as float because it contains missing values; convert it back to Pandas nullable Int64 when needed.

6. Outlier Analysis

Potential outliers in age, quantity and amount were reviewed using the Interquartile Range (IQR) method and box plots. An IQR flag only identifies an unusual value; it does not prove that the record is wrong.

Column	Q1	Q3	IQR	IQR limits	Flagged	Decision
Age	28	47	19	-0.5 to 75.5	600 (1.93%)	Retained; ages 76-78 may be valid.
Quantity	1	1	0	1 to 1	171 (0.55%)	Retained; IQR is zero because most records have quantity 1.
Amount	Rs. 486	Rs. 794	Rs. 308	Rs. 24 to Rs. 1,256	1,115 (3.59%)	Retained; higher-value purchases may be genuine.

How to read these results

Age: The statistical upper limit is 75.5, so ages 76-78 are flagged even though they may be perfectly valid customer ages.

Quantity: Q1 and Q3 are both 1, giving an IQR of zero. This makes the IQR rule flag values above 1; it is not a sensible reason by itself to remove multi-item purchases.

Amount: Some records exceed the upper IQR limit of Rs. 1,256. They were kept because a higher transaction amount can reflect a genuine purchase.

Outlier decision

No IQR-flagged values were removed. The flagged records were retained because the available checks did not establish that they were data-entry errors. They can be investigated further if product-level or transaction-level business rules become available.

7. Final Validation Checklist

Validation area	Result	What it means
Dataset structure	31,047 x 21	Final shape confirmed.
Exact duplicate rows	0 in final data	No exact duplicates remained; before-cleaning duplicate count was not supplied.
Missing values	2,084 cells across 5 columns	Remaining gaps are documented; no unsupported identifiers or dates were invented.
Numeric validity	Basic ranges passed	Age 18-78, quantity 1-20 and amount Rs. 229-Rs. 3,036; no zero / negative quantity or amount.
Category consistency	Standardized	Common text labels, city labels and state labels were standardized.
Date validity	Partially checked	Valid dates were parsed; 488 dates remain unavailable.
Outliers	Reviewed and retained	IQR flags were not treated as errors without evidence.

8. Remaining Issues and Recommended Next Checks

1. Trace the 1,080 missing order IDs back to the source system if order-level analysis requires them; do not create replacement IDs.

2. Review the 41 remaining missing customer IDs and the 488 missing dates. Derive month only where a valid date is available.

3. Review the 465 missing month values and keep them separate from known months in time-based analysis.

4. Validate postal codes against an authoritative reference if accurate location-level reporting is required.

5. Add explicit checks for future dates, allowed category values, ID formats, whitespace-only strings, and raw Boolean representations.

6. After CSV export and reload, parse the date column and restore postal-code dtype to nullable Int64 if needed.

7. Keep the original raw dataset unchanged and store the cleaned dataset separately.

9. Conclusion

The SwiftCart dataset was cleaned and checked for the main data quality issues identified during the project. The final dataset has 31,047 rows and 21 columns, with no exact duplicate rows and valid basic ranges for age, quantity and amount. Text categories and date formats were standardized where possible, and potential outliers were retained when there was no evidence that they were incorrect.

Final assessment: The dataset is ready for exploratory analysis, provided the remaining missing IDs, dates, month values and postal codes are considered when interpreting results. This is a practical validation of the checks performed, not a guarantee that every value has been independently verified against the original retail system.

Report basis: SwiftCart cleaning and validation outputs shared during the project workflow. Final values are reported as observed; items without supporting outputs are identified as limitations rather than assumed to have passed.